WMO Activities in support on Climate Data

Contribution from SERCOM Expert Teams
Standing Committee on Climate Services

Teams:

ET-DDS - Expert Team on Data Development and Stewardship

ET-MCCVC - Expert Team on Monitoring and Communications of Climate Variability and change

Speakers:

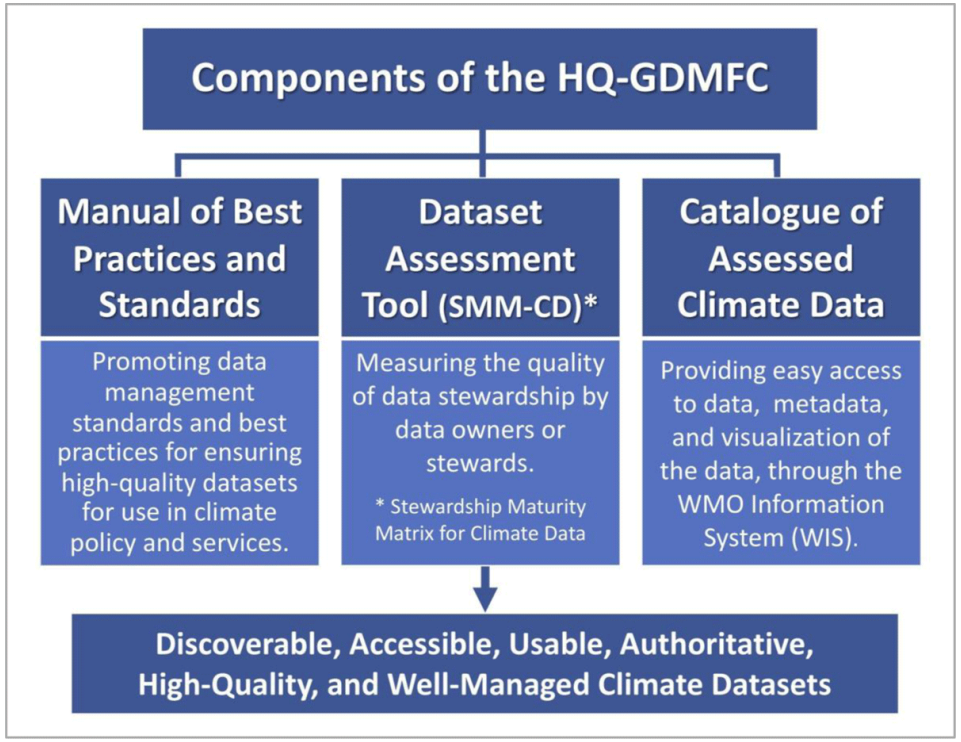
Robert Dunn – UK Met Office (**ET-DDS**)
Blair Trewin – Bureau of Meteorology, Australia (**ET-MCCVC**)

…Supporting Members in delivering high-quality, consistent and usable climate data

WORLD METEOROLOGICAL ORGANIZATION

Data Rescue, Generation, and Homogenization: Core Priorities for WMO Climate Data

Recognised as key components in the WMO Climate Data Framework
Strengthened in:
WMO-No. 1238 - Manual on the High-quality Global Data Management Framework for Climate
WMO-No. 1131 - Climate Data Management System Specifications



Data Rescue
- Preservation and digitization of historical climate records
- Essential for extending and enhancing long-term climate series
- Increasingly embedded in national strategies and WMO guidance

Climate Dataset Generation
- Climate Observations, Data Exchange
- QC, Basic statistics
- Data Stewardship

Homogenization
- Ensures consistency and comparability of climate time series
- Critical for climate monitoring, trends and indices
- Requires sustained methodological coordination across Members

Together, Data Rescue and Homogenization form the foundation of reliable, long-term climate datasets.

Engaging Members : Consultations on key domains 1/2

Consultation on Homogenization (2025)

- Global consultation held in November 2025
- Two sessions, multiple languages
- Participation from:
  - Homogenization experts
  - Regional Climate Centres use cases (South America)
  - NMHSs use cases (Madagascar, Hungary)
- Survey on homogenization practices

Results in the following presentation (Jose Guijarro)

Engaging Members: Dataset Maturity Assessment 2/2

Upcoming consultation on the WMO Stewardship Maturity Matrix for Climate Data [SMM-CD] (WMO-No. 1328)

- Enables NMHSs and RCCs to:
  - Self-assess climate datasets
  - Improve transparency and usability
  - Provide consistent information to users

Dunn et al, 2021

Towards harmonization

- Four key dimensions:
  - Data Access
  - Usability and usage
  - Quality Management
  - Data Management

	→ SMM-CD Category →			
↓ Aspect ↓	**Data Access**	**Usability & Usage**	**Quality Management**	**Data Management**
	Discoverability	Data Portability	Quality Assurance & Control Procedure	Preservation
	Accessibility	Documentation	Quality Assessment	Metadata
		Usage & Impact	Data Integrity	Governance

Need to harmonize existing maturity approaches across regions and organizations

DAYCLI: Standardizing the Exchange of Daily Climate Data 1/2

What is DAYCLI?

- WMO Standard for the **monthly** exchange of **daily climate data**
- Covers key variables:
  - Temperature (min, max, mean)
  - Precipitation
  - Snow depth (fresh and total)
- Supports climate **monitoring, assessment, and services**

Key Principles

- Exchange of **validated, nationally consistent datasets**
- Derived from CDMS workflows (not raw observations)
- Based on **national climatological practices**
- Includes **metadata and quality information**

```
wsi_series,wsi_issuer,wsi_issue_number,wsi_local,wmo_block_number,wmo_station_numbe
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,1,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,2,0,7,0,1,1.5,0,0,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,3,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,4,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,5,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,6,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,7,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,8,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,9,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,10,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,11,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,12,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,13,0,7,0,1,0,0,0,7,
0,20000,0,87765,87,765,-41.1475,-71.16417,835,255,255,2,2021,11,14,0,7,0,1,0,0,0,7,
```

DAYCLI: Standardising the Exchange of Daily Climate Data 2/2

DAYCLI complements existing WMO Data Exchange:

- Real-time observations
- Monthly CLIMAT messages

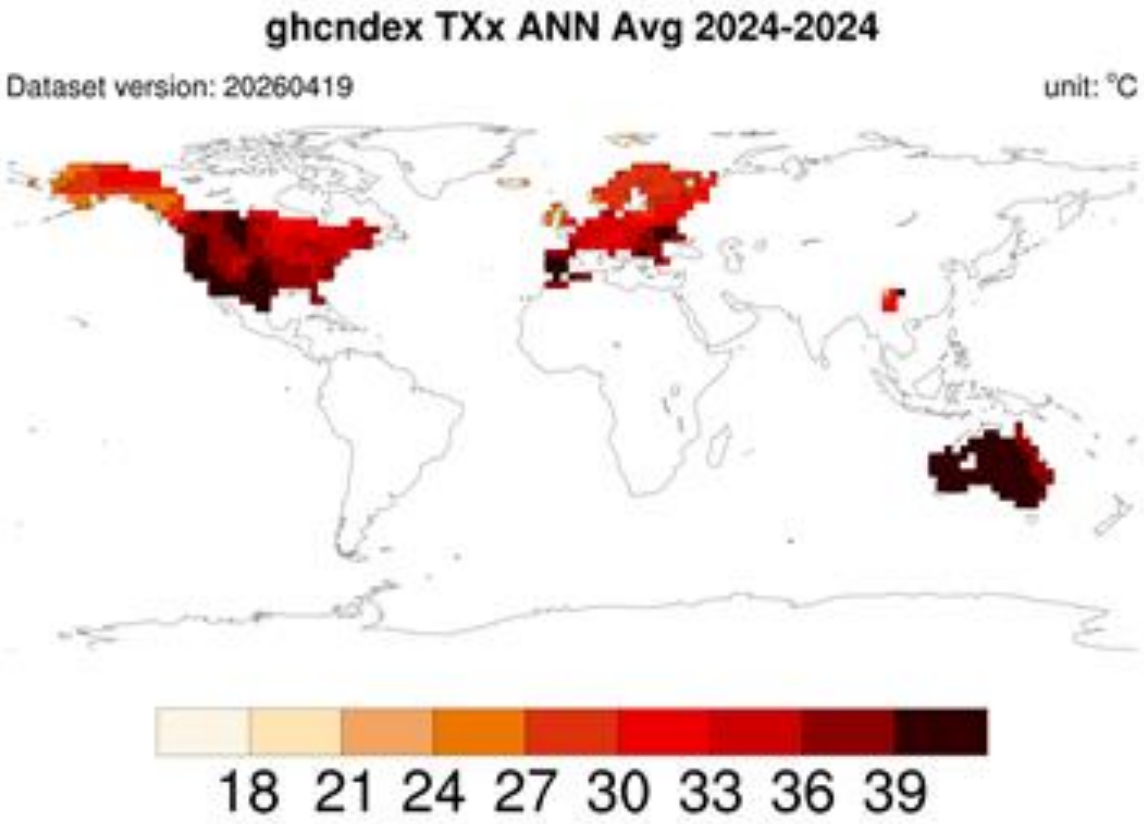
→ **Provides a standardized framework for daily climate data exchange**

2026 update: Key improvements:

- Reinforces exchange of quality-controlled data
- Preserves long-term time series consistency
- Aligns with NMHS operational practices
- Reduces ambiguity in time references
- Improves global comparability and reliability

→ **DAYCLI bridges the gap between real-time observations and climate data stewardship**

Lack of accessible daily data limits our capacity to monitor climate extremes



copyright www.climdex.org, 2026-05-04
10.1175/BAMS-D-12-00109.1

Climate indices: Towards a New Generation of Climate Metrics 1/2

Led by ET-MCCVC

https://www.climdex.org/learn/indices /

Why a review is needed

- Evolving user needs and decision-making contexts
- Increasing focus on extremes and impacts
- Emergence of new variables (e.g. heat stress, humid heat)
- Expansion to new domains (e.g. marine, hydrology)

Key Areas of Work

- Review and update existing indices (ETCCDI, ET-SCI)
- Explore new variables and timescales (including sub-daily)
- Assess availability and sustainability of datasets
- Strengthen links between indices and user needs
- Evaluate tools, software, and governance frameworks

Heat and cold

FD, SU, ID, TR, GSL, TXx, TNx, TXn, TNn, TN10p, TX10p, TN90p, TX90p, WSDI, CSDI, DTR, ETR, CDDcoldn, GDDgrown, HDDheatn, TMge5, TMlt5, TMge10, TMlt10, TMm, TXm, TNm, TXge30, TXge35, TXgt50p, TNlt2, TNltm2, TNltm20, TXbdTNbd

Heat and cold waves

HWN, HWF, HWD, HWM, HWA, CWN_ECF, CWF_ECF, CWD_ECF, CWM_ECF, CWA_ECF

Precipitation

Rx1day, Rx5day, SPI, SPEI, SDII, R10mm, R20mm, Rnnmm, CDD, CWD, R95p, R99p, R95pTOT, R99pTOT, PRCPTOT

WORLD METEOROLOGICAL ORGANIZATION

Climate indices: Towards a New Generation of Climate Metrics 2/2

Strategic Direction

- Move towards a **more flexible and user-oriented framework**
- Consider **tiered indices** (different levels of priority)
- Improve coordination on **data exchange and licensing**
- Ensure long-term institutional support

Monitoring Outputs

- Refreshed and revised indices will support **monitoring activities**

This work will define the next generation of climate indices, supporting both scientific analysis and operational climate services.

Strengthening Climate Data for Services and Decision-Making

High-quality climate data are the foundation of climate understanding, monitoring and services

No data → No understanding
No data → No services

WMO addresses climate data challenges through coordinated global action, with SERCOM Expert Teams playing a central role:

- **ET-DDS** – Standardisation of climate information, climatological practices, Data Rescue and Scientific methods (e.g. homogenisation, datasets assessment)
- **ET-MCCVC** - Monitoring, climate indices, communication of variability and change

Together, these activities ensure that climate data are:

- Accessible
- Reliable
- Consistent
- Fit for purpose

Reliable climate services start with reliable climate data

WORLD METEOROLOGICAL ORGANIZATION

WMO Expert Teams Supporting Climate Data

ET-DDS Co-Chairs

- Denis Stuber
  denis.stuber@meteo.fr
- Reinaldo Silviera
  reinaldo.silveira@simepar.br

ET-MCCVC Co-Chairs

- Blair Trewin
  Blair.Trewin@bom.gov.au
- Jessica Blunden

WMO Secretariat: Peer Hechler – phechler@wmo.int

WORLD METEOROLOGICAL ORGANIZATION